

1) Explain in brief Data Science with examples

Answer:

1. Data science is an interdisciplinary field that focuses on extracting meaningful information and knowledge from large volumes of data.
2. It combines techniques from statistics, machine learning, visualization and computing to analyze data systematically.
3. The primary objective of data science is to discover patterns and build predictive models that support decision making.
4. Data science works with both structured data like databases and unstructured data such as text, images and videos.
5. Organizations use data science to automate decision processes and improve efficiency in operations.
6. Examples include fraud detection in banking, movie recommendations in Netflix and customer churn prediction in business.

 **Reference:** Data Science Concepts and Practice, Ch-1 p.1-4

 Data-Science-Concepts-and-Pract...

2) List and elaborate Data Science tasks and examples

Answer:

1. Classification is a task used to predict categorical outcomes such as deciding whether a loan should be approved or rejected.
2. Regression is used to predict numeric values like estimating house price or forecasting sales revenue.
3. Clustering groups similar data records together, for example segmenting customers based on purchasing behavior.
4. Association analysis identifies relationships between items, such as products frequently bought together in a supermarket.
5. Anomaly detection identifies unusual patterns in data, which is commonly used in credit card fraud detection.
6. Time series forecasting predicts future values based on past data such as predicting next month's temperature or sales.

 **Reference:** Data Science Concepts and Practice, p.10-11

 Data-Science-Concepts-and-Pract...

3) Discuss CRISP-DM

Answer:

1. CRISP-DM is a standard framework that describes the steps involved in solving a data science problem.
2. The first step is business understanding where the objective and problem statement are clearly defined.
3. The second step is data preparation where relevant data is collected, cleaned and transformed.
4. The third step is modeling where suitable algorithms are applied to build predictive models.
5. The fourth step is evaluation where the model is tested using real-world data to check accuracy.
6. The final step is deployment where the model is implemented and continuously maintained in practice.

 **Reference:** ADS Module 1 PPT – Data Science Process

 ADS module 1

4) Differentiate Causation vs Correlation

Causation	Correlation 
Causation means one variable directly produces an effect on another variable.	Correlation means two variables change together but not necessarily due to cause.
It represents a cause-and-effect relationship between variables.	It represents only statistical association between variables.
If A causes B, then changing A will directly influence B.	Changing one variable does not guarantee change in the other.
Used for explanation and decision making.	Used mainly for measuring relationship strength.
Example: Smoking causes lung disease.	Example: Ice cream sales increase with temperature.

 **Reference:** ADS Module 1 PPT – Causation vs Correlation

 ADS module 1

5) Interval Scale vs Ratio Scale

Interval Scale	Ratio Scale
Interval scale has equal intervals but no true zero point.	Ratio scale has equal intervals and an absolute zero.
Differences between values are meaningful.	Both differences and ratios are meaningful.
We cannot say one value is twice another value.	We can say one value is twice another value.
Used mainly for temperature in Celsius or Fahrenheit.	Used for height, weight and income measurements.
Supports addition and subtraction operations only.	Supports all mathematical operations.

 **Reference:** Data Exploration PPT – Numeric attribute types

 Data Exploration-1



6) Discrete vs Continuous

Discrete Data	Continuous Data
Discrete data consists of countable values.	Continuous data consists of measurable values.
Values are usually integers.	Values can be decimal or fractional numbers.
Obtained through counting process.	Obtained through measurement process.
Example: number of students in a class.	Example: height and temperature.
Cannot take infinite values in a range.	Can take infinite values within a range.

 **Reference:** Data Exploration PPT – Discrete vs Continuous  Data Exploration-1

7) Why is it important to measure the spread of data?

Answer:

1. Measuring spread helps to understand how far the data values are distributed from the average value.
2. It determines whether the mean properly represents the dataset.
3. A large spread indicates high variability among observations.
4. A small spread shows that data values are close to each other.
5. Spread measures help in detecting outliers and unusual observations.
6. It supports statistical comparison and better decision making.

 **Reference:** Data Exploration PPT – Measure of spread  Data Exploration-1

8) Discuss Pearson's first and second coefficient

Answer:

1. Pearson's coefficients are used to measure the skewness of a distribution.
2. The first coefficient uses mean, mode and standard deviation.
$$Sk_1 = (\text{Mean} - \text{Mode}) / \text{Standard deviation}$$
3. The second coefficient uses mean, median and standard deviation.
$$Sk_2 = 3(\text{Mean} - \text{Median}) / \text{Standard deviation}$$
4. Positive value indicates positively skewed distribution.
5. Negative value indicates negatively skewed distribution.
6. Zero value indicates symmetrical distribution.

 **Reference:** Data Exploration PPT – Skewness  Data Exploration-1

9) Give two applications of skewness

Answer:

1. Skewness is used in income distribution analysis to identify inequality in population earnings.
2. It helps in analyzing exam result distributions to understand difficulty level of the test.
3. It is useful in medical research for analyzing patient measurement distributions.
4. It helps in cricket score analysis to evaluate player performance trends.
5. It is applied in finance to study stock return behavior.
6. It helps identify presence of extreme values in datasets.

 **Reference:** Data Exploration PPT – Skewness examples  Data Exploration-1

10) State and explain Standard Deviation with formula

Answer:

1. Standard deviation measures the average distance of data values from the mean.
2. It indicates how much variation exists in the dataset.
3. Small standard deviation means data values are close to mean.
4. Large standard deviation means data values are widely scattered.
5. It is the square root of variance and always non-negative.
6. Formula:

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

 **Reference:** Data Exploration PPT – Standard deviation

 Data Exploration-1

11) Explain Quartile and IQR

Answer:

1. Quartiles divide a dataset into four equal parts based on position.
2. First quartile (Q1) represents 25% of observations.
3. Second quartile (Q2) represents median of the dataset.
4. Third quartile (Q3) represents 75% of observations.
5. Interquartile range is calculated as Q3 minus Q1.
6. IQR measures the spread of middle values and reduces effect of outliers.

 **Reference:** Data Exploration PPT – Quartiles and IQR

 Data Exploration-1

12) Absolute deviation and Variance

Answer:

1. Absolute deviation measures the average distance of each value from the mean.
2. Mean absolute deviation is obtained by averaging these deviations.
3. Variance measures squared deviations of values from mean.
4. Larger variance indicates greater variability in data.
5. Smaller variance indicates data values are closely grouped.
6. Standard deviation is the square root of variance.

 **Reference:** Data Exploration PPT – Absolute deviation & variance

 Data Exploration-1

1) Explain the Data Science Process

Answer:

The data science process is a systematic and iterative approach used to extract knowledge from data and solve real-world problems.

Steps in Data Science Process

1. Business Understanding (Problem Definition)

- The first step is to clearly understand the objective or problem that needs to be solved.
- The analyst identifies what decision needs support and what data is required.
- A wrong problem statement leads to incorrect model results.

2. Data Collection & Understanding

- Relevant data is collected from databases, sensors, web sources or logs.
- The analyst studies attributes, relationships and structure of the dataset.
- Domain knowledge is important to interpret data correctly.

3. Data Preparation

- Raw data is cleaned and transformed into suitable format.
- Missing values, duplicates and outliers are handled.
- Data is converted into structured tabular form for modeling.

4. Modeling

- Machine learning algorithms are applied to build predictive models.
- The model learns patterns from training data.
- Different models are tested and compared.

5. Evaluation

- Model accuracy and performance are checked using test data.
- If results are unsatisfactory, earlier steps are repeated.

6. Deployment

- Final model is implemented in real applications.
- The system is monitored and updated periodically.

➡ The process is iterative, meaning steps may repeat until acceptable performance is achieved.

2) Discuss in detail Data Preparation

Answer:

Data preparation is the most time-consuming phase of data science where raw data is converted into usable format.

Major Activities in Data Preparation

1. Data Exploration

- Statistical summaries and visualization are used to understand data patterns.
- Helps identify distribution, skewness and correlations.

2. Data Cleaning (Data Quality)

- Duplicate records are removed.
- Incorrect or inconsistent values are corrected.

3. Handling Missing Values

- Missing values are replaced using mean, median or predictive techniques.
- Understanding the reason for missing data is important before replacement.

4. Data Transformation

- Data is converted into required format using normalization or scaling.
- Categorical data may be encoded into numeric form.

5. Outlier Detection

- Extreme values that distort analysis are identified and treated.
- Outliers can affect mean and regression models.

6. Feature Selection

- Only important attributes are selected to improve model accuracy.
- Reduces dimensionality and improves efficiency.

7. Sampling

- Large datasets may be sampled to reduce computational cost.

3) Explain Modeling in Data Science Process

Answer:

Modeling is the stage where algorithms learn patterns from prepared data to make predictions.

Steps in Modeling

1. Selecting Algorithm

- Choice depends on problem type: classification, regression or clustering.
- Examples: Decision tree, regression, neural network.

2. Training the Model

- Training dataset with known outputs is provided.
- The algorithm learns relationship between input and output variables.

3. Parameter Tuning

- Model parameters are adjusted to improve accuracy.
- Prevents overfitting and underfitting.

4. Testing the Model

- Test dataset is used to evaluate predictive performance.
- Accuracy, precision and recall are measured.

5. Model Comparison

- Multiple models are compared to choose the best one.
- Best model balances accuracy and complexity.

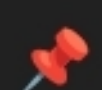
6. Final Model Selection

- Selected model is used for deployment.

 **Reference:** Data Science Concepts and Practice – Model building concept p.5

4) Difference between Data Science and Data Analytics

Basis	Data Science	Data Analytics
Definition	Data science is a broad interdisciplinary field that extracts knowledge from data and builds predictive intelligent systems.	Data analytics is the process of examining datasets to understand patterns and generate useful reports.
Objective	Focuses on predicting future outcomes and automating decision making.	Focuses on understanding past and present data trends.
Nature of Work	Exploratory, predictive and prescriptive in nature.	Mainly descriptive and diagnostic in nature.
Techniques Used	Uses machine learning, artificial intelligence, statistics and advanced algorithms.	Uses statistical analysis, querying, aggregation and visualization tools.
Type of Data	Works with structured, semi-structured and unstructured data such as text, images and videos.	Mostly works with structured data from databases and spreadsheets.
Output	Produces predictive models and intelligent systems.	Produces dashboards, charts and reports.
Programming Requirement	Requires programming knowledge (Python, R, ML frameworks).	Often uses tools like Excel, SQL, Power BI, Tableau with less programming.
Decision Making	Supports automated decision making systems.	Supports human decision making by providing insights.
Example	Fraud detection, recommendation system, demand forecasting.	Monthly sales analysis, customer trend report.
Skill Set	Requires knowledge of statistics, programming, machine learning and domain expertise.	Requires knowledge of statistics, business understanding and visualization.



6) Describe Measure of Central Tendency with example

Answer:

Measures of central tendency are statistical measures that represent the central or typical value of a dataset. They help summarize large datasets into a single representative value.

Types of Central Tendency

1. Mean (Arithmetic Average)

- Mean is calculated by dividing the sum of all observations by total number of observations.

$$Mean = \frac{\sum x}{n}$$

- It considers every value in dataset.
- It is affected by extreme values (outliers).

2. Median (Middle Value)

- Median is the middle value when data is arranged in ascending or descending order.
- It divides the dataset into two equal halves.
- It is preferred when data contains outliers.

3. Mode (Most Frequent Value)

- Mode is the value that appears most frequently in dataset.
- It is useful for categorical data.

Example

Data: 2, 4, 6, 8, 20

Mean = 8 (affected by extreme value 20)

Median = 6 (middle value)

Mode = none

This shows mean may misrepresent skewed data while median gives better central value.

8) Explain Skewness and types of skewness

Answer:

Skewness is a statistical measure that describes the asymmetry of a data distribution around its mean. A distribution is symmetric if both sides of the curve are identical. When one side is longer, the distribution becomes skewed.

Types of Skewness

1. Positive Skewness (Right Skewed)

- The tail of distribution extends towards the right side.
- Majority of values are small and few values are extremely large.
- $\text{Mean} > \text{Median} > \text{Mode}$
- Example: income distribution where few people earn very high salaries.

2. Negative Skewness (Left Skewed)

- The tail extends towards the left side.
- Majority of values are high and few values are extremely small.
- $\text{Mean} < \text{Median} < \text{Mode}$
- Example: marks in an easy exam where most students score high.

3. Symmetrical Distribution

- Both sides of distribution are equal.
- $\text{Mean} = \text{Median} = \text{Mode}$
- Example: normal distribution of heights.

Skewness helps identify data bias and suitability of statistical methods.

9) Explain Kurtosis with formula and example

Answer:

Kurtosis is a statistical measure used to describe the shape of distribution in terms of peakedness or flatness compared to a normal distribution.

Formula

$$Kurtosis = \frac{E[(X - \mu)^4]}{\sigma^4}$$

Where

μ = mean

σ = standard deviation

Types of Kurtosis

1. Leptokurtic

- Distribution has a sharp peak and heavy tails.
- Contains many outliers.
- Example: stock market returns

2. Mesokurtic

- Normal distribution shape.
- Moderate peak and moderate tails.

3. Platykurtic

- Flat peak and thin tails.
- Fewer extreme values.
- Example: uniform distribution data

Kurtosis helps identify presence of extreme values and risk in data analysis.

10) Explain Correlation with suitable example

Answer:

Correlation is a statistical measure that determines the strength and direction of relationship between two variables. It shows how change in one variable affects another variable.

Formula

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sigma_x \sigma_y}$$

Where r ranges from -1 to +1.

Types of Correlation

Positive Correlation

- Both variables increase or decrease together.
- Example: study hours and exam marks.

Negative Correlation

- One variable increases while the other decreases.
- Example: speed and travel time.

Zero Correlation

- No relationship between variables.
- Example: shoe size and intelligence.

Correlation helps in prediction but does not prove cause-effect relationship.